SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I __Keerthana R__(192524489) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: KEERTHANA R

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

ANSWER:

Given:

Allocated IP Address Block: 192.168.100.0/24

Subnet Allocation:

- School of Computing → **/25**
- School of Electronics → **/26**
- School of Mechanical Engineering → **/27**

The subnet allocation is performed from the largest subnet to the smallest subnet.

1. Determine the Network Address and Broadcast Address

School of Computing (/25)

Network Address: **192.168.100.0**

Broadcast Address: **192.168.100.127**

School of Electronics (/26)

Network Address: **192.168.100.128**

Broadcast Address: **192.168.100.191**

School of Mechanical Engineering (/27)

Network Address: **192.168.100.192**

Broadcast Address: **192.168.100.223**

2. Identify the Valid Host IP Address Range

School of Computing (/25)

First Host Address: **192.168.100.1**

Last Host Address: **192.168.100.126**

Valid Host Range: **192.168.100.1 – 192.168.100.126**

School of Electronics (/26)

First Host Address: **192.168.100.129**

Last Host Address: **192.168.100.190**

Valid Host Range: **192.168.100.129 – 192.168.100.190**

School of Mechanical Engineering (/27)

First Host Address: **192.168.100.193**

Last Host Address: **192.168.100.222**

Valid Host Range: **192.168.100.193 – 192.168.100.222**

3. Calculate the Number of Usable Host Addresses

The number of usable host addresses is calculated using the formula:

Usable Hosts = 2^(Host Bits) − 2

School of Computing (/25)

Host Bits = 7

Total Addresses = 2^7 = 128

Usable Hosts = **128 − 2 = 126**

School of Electronics (/26)

Host Bits = 6

Total Addresses = 2^6 = 64

Usable Hosts = **64 − 2 = 62**

School of Mechanical Engineering (/27)

Host Bits = 5

Total Addresses = 2^5 = 32

Usable Hosts = **32 − 2 = 30**

4. Determine the Remaining Unused IP Address Range

The allocated subnets occupy the following ranges:

School of Computing: **192.168.100.0 – 192.168.100.127**

School of Electronics: **192.168.100.128 – 192.168.100.191**

School of Mechanical Engineering: **192.168.100.192 – 192.168.100.223**

Therefore, the remaining unused IP address range is:

Network Address: 192.168.100.224

First Host Address: 192.168.100.225

Last Host Address: 192.168.100.254

Broadcast Address: 192.168.100.255

This remaining block is equivalent to the subnet **192.168.100.224/27**, which provides **30 usable host addresses**.

1. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWER:

Given:

Allocated IP Address Block: 10.10.0.0/16

Subnet Allocation:

- Software Development → **/24**
- Quality Assurance → **/25**
- Human Resources → **/26**
- Executive Management → **/27**

The subnets are allocated from the largest subnet to the smallest subnet.

1. Determine the Network Address and Broadcast Address

Software Development (/24)

Network Address: 10.10.0.0

Broadcast Address: 10.10.0.255

Quality Assurance (/25)

Network Address: 10.10.1.0

Broadcast Address: 10.10.1.127

Human Resources (/26)

Network Address: 10.10.1.128

Broadcast Address: 10.10.1.191

Executive Management (/27)

Network Address: 10.10.1.192

Broadcast Address: 10.10.1.223

2. Identify the Valid Range of Host IP Addresses

Software Development (/24)

First Host Address: 10.10.0.1

Last Host Address: 10.10.0.254

Valid Host Range: 10.10.0.1 – 10.10.0.254

Quality Assurance (/25)

First Host Address: 10.10.1.1

Last Host Address: 10.10.1.126

Valid Host Range: 10.10.1.1 – 10.10.1.126

Human Resources (/26)

First Host Address: 10.10.1.129

Last Host Address: 10.10.1.190

Valid Host Range: 10.10.1.129 – 10.10.1.190

Executive Management (/27)

First Host Address: 10.10.1.193

Last Host Address: 10.10.1.222

Valid Host Range: 10.10.1.193 – 10.10.1.222

3. Calculate the Number of Usable Hosts

The number of usable host addresses is calculated using the formula:

Usable Hosts = 2^(Host Bits) – 2

Software Development (/24)

Host Bits = 8

Total Addresses = 2^8 = 256

Usable Hosts = 254

Quality Assurance (/25)

Host Bits = 7

Total Addresses = 2^7 = 128

Usable Hosts = 126

Human Resources (/26)

Host Bits = 6

Total Addresses = 2^6 = 64

Usable Hosts = 62

Executive Management (/27)

Host Bits = 5

Total Addresses = 2^5 = 32

Usable Hosts = 30

4. Identify the Unused IP Address Range

The allocated subnets occupy the following ranges:

Software Development: **10.10.0.0 – 10.10.0.255**

Quality Assurance: **10.10.1.0 – 10.10.1.127**

Human Resources: **10.10.1.128 – 10.10.1.191**

Executive Management: **10.10.1.192 – 10.10.1.223**

The remaining unused IP address range within the allocated **10.10.0.0/16** block is:

Start Address: **10.10.1.224**

End Address: **10.10.255.255**

This remaining address space is available for future subnet allocation and network expansion.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER:

Given:

IPv6 Network Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Prefix Length: /64

1. Convert the Given IPv6 Address into Compressed Notation

IPv6 compression rules:

Remove leading zeros in each hexadecimal block.

Replace one consecutive sequence of all-zero blocks with ::.

Given Address:

2001:0db8:abcd:0000:0000:0000:0000:0000

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the Compressed Address Back into Full Hexadecimal Representation

Compressed Address:

2001:db8:abcd::

Expanding the omitted zero blocks gives:

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

3. Identify the Network Prefix Using the /64 Prefix Length

A **/64 prefix** means:

- The **first 64 bits** represent the **network prefix**.
- The **last 64 bits** represent the **interface identifier (host portion)**.

Network Prefix:

2001:0db8:abcd:0000::/64

or simply,

2001:db8:abcd:0::/64

4. Calculate the Total Number of Host Addresses Supported Within the Subnet

IPv6 uses **128-bit addresses**.

Host bits = **128 − 64 = 64**

Number of host addresses:

2^{64} = 18,446,744,073,709,551,616

Therefore, the subnet supports:

18,446,744,073,709,551,616 host addresses

5. Total Number of Possible Host Addresses Available Within This Network

Since the network prefix is **/64**, the host portion contains **64 bits**.

Therefore,

Total Possible Host Addresses = 2^{64}

= **18,446,744,073,709,551,616**

Unlike IPv4, **IPv6 does not reserve network and broadcast addresses**, so all **2^{64} addresses are available** within the subnet.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry** .
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

ANSWER:

Given:

The router has the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP Address: 192.168.2.45

1. Determine the Destination Subnet

The subnet mask **/24** corresponds to **255.255.255.0** .

Applying the subnet mask:

Destination IP: **192.168.2.45**

Subnet Mask: **255.255.255.0**

Network Address = 192.168.2.0

Therefore, the packet belongs to the subnet:

192.168.2.0/24

2. Identify the Matching Routing Table Entry

Compare the destination network with the routing table:

Routing Table Entry	Match
192.168.1.0/24	No
192.168.2.0/24	**Yes**
192.168.3.0/24	No

Matching Routing Table Entry:
192.168.2.0/24

3. Determine the Next-Hop Network/Interface

Since the destination IP address belongs to the subnet **192.168.2.0/24**, the router forwards the packet through the interface or next-hop connected to this network.

Next-Hop Network/Interface:
Interface connected to the 192.168.2.0/24 network (e.g., Interface G0/1 or the configured next-hop for 192.168.2.0/24).

4. Default Route if No Match Exists

If the destination IP address does not match any routing table entry, the router forwards the packet using the **default route**.

The default route in IPv4 is:

0.0.0.0/0

This route directs packets to the default gateway or ISP when no specific route is found

5.A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for

communication.

ANSWER:

Given:

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Router Interface (Default Gateway):

Administration Gateway: 192.168.10.1

Finance Gateway: 192.168.10.65

Subnet Mask for /26: 255.255.255.192

1. Determine the Network Address and Broadcast Address

Administration LAN (192.168.10.0/26)

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Finance LAN (192.168.10.64/26)

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

2. Identify the Valid Host IP Address Range

Administration LAN

First Host Address: 192.168.10.1

Last Host Address: 192.168.10.62

Valid Host Range: 192.168.10.1 – 192.168.10.62

Finance LAN

First Host Address: 192.168.10.65

Last Host Address: 192.168.10.126

Valid Host Range: 192.168.10.65 – 192.168.10.126

3. Suggest Suitable IP Addresses for Three Computers in Each LAN

Administration LAN

Computer	**IP Address**
Computer 1	192.168.10.2

Computer 2	192.168.10.3

Finance LAN

Computer	IP Address
Computer 1	192.168.10.66
Computer 2	192.168.10.67
Computer 3	192.168.10.68

4. Identify the Default Gateway for Each LAN

Administration LAN

Default Gateway: 192.168.10.1

Finance LAN

Default Gateway: 192.168.10.65

The default gateway is the router interface through which hosts communicate with devices outside their local network.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand